MEDICAL TECHNOLOGY LICENSURE EXAM REVIEW - **HTMLE**

LABORATORY MANAGEMENT & OPERATIONS

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Notes by: Xiao - The Conqueror of Demons, The Vigilant Yaksha, & Alatus, the Golden-Winged King

LABORATORY MANAGEMENT

- Over-all handling of the Clinical Laboratory
- Requires LEADER - provide direction, and MANAGER to get things done
- Organization – good only as its people – guided by **LEADERS** and **MANAGEMENT**
- Qualities of Leader and Manager

LEADER	MANAGER
Administrator	Implementer
Organizer and Developer	Watches Bottom Line
Risk-Taker	Maintains Control
Inspiration	Is Good Soldier
Thinks Long Term	Thinks Short Term
Asks WHAT and WHY	Asks HOW and WHEN
Challenges status quo	Accepts status quo
Does the Right Thing	Does Things Right

- Administrator
  - Assign and perform tasks
  - Coach team members
- Organizer and Developer
  - Transform an ordinary work to an extraordinary one
  - Helps colleague succeed
- Maintains Control
  - "I'm the boss, you have to follow what I'm going to say"

LEADERSHIP

- Pattern of behaviors used to engage other to complete tasks in a *timely and productive manner*
  - Timely and productive: employee is learning while doing the tasks
- Leadership Styles:
  - Supportive Leader: helps in doing the task
  - Directive Leader: instructs / follows up
  - Delegating Leader: believes that if work is distributed, it will be done in a timely and productive manner
  - Coaching Leader: teaches the step-by-step procedure, demonstrates

MANAGEMENT

- Uses the *human, financial, physical, and information resources* available to an organization in the **MOST EFFICIENT** and **EFFECTIVE** manner
- Types of Management:
  - **Operations Management**
    - Management of day to day operations
    - From beginning to the last hour of shift
  - **Human Resource Management**
    - In-charge of recruitment
    - Develops and evaluates employee
    - Provides training
  - **Financial Management**
  - **Marketing Management**
    - Most important
    - Effective marketing is needed for a product to be sold

STRATEGIC PLANNING

- Continuous **REASSESSMENT** of goals and services and adjust to market forces like staffing shortage, stricter regulatory mandates, and new sophisticated technologies to survive competitive settings.
  - Reassessment: re-evaluate plan from time to time = is output good/ bad? → if bad → plan again
- Techniques in Strategic Planning:
  - Histogram/Graph/Scattergram
  - Brainstorming
  - Fishbone Diagram: cause and effect relationship
  - Storyboarding: timeline
  - Analysis: Pareto, Delphi, SWOT

I. LABORATORY QUALITY

- Accuracy, Reliability, Timeliness of Reported Results (according to CLSI)
  - Quality is an A-R-T
- Consequences of Laboratory Errors
  - Unnecessary Treatment
  - Treatment Complications
  - Delay in Correct Diagnosis
  - Additional and Unnecessary Diagnostic Test
- **Quality Management System**: coordinated activities to direct and control an organization with regard to quality.
- Paths of Workflow
  - entire set of operations that occur in testing begins with the **PATIENT**
  - ends with the **REPORTING, INTERPRETATION**

- Overview of Quality Management System Model
  - Organized (understandable, workable) Structure
  - Ensures appropriate management
  - Model Developed by CLSI
  - Compatible with ISO Standard
  - Quality Manager

QUICK CONCEPTS:

TERM	DESCRIPTION
Organization	• Structure and management of the Laboratory • Establishment of Policies, Implementation, Monitoring
Personnel	• Most important • Competent and motivated (excellent productivity / output) • Encouragement and Motivation
Equipment	• Choose the *RIGHT EQUIPMENT* (1 machine for multiple tests, follow minimum dimension), Proper Maintenance
Purchasing and Inventory	• Manage Reagents and Supplies; Cost Savings; Ensure availability of supplies • Quality organization = ability to follow minimum standard
Process Control	• Quality Control, Sample Management, Verification and Validation of Methods
Information Management	• Product of Laboratory: *INFORMATION* • Accuracy, Confidentiality, Accessibility
Documents	• Inform how to do things
Records	• Meticulously maintained: accurate and accessible
Occurrence Management	• "occurrence" – an error that should not have happened • Handle, Learn, Take Action= NOT HAPPEN AGAIN
Assessment	• Tool to examine: *PERFORMANCE* (machine, personnel, system) • Benchmark (copying practices which are efficient / helpful); • Internal and External Assessment Tool
Process Involvement	• Systematic Process Improvement
Customer Service	• Understand who the customers are. • Assess needs and feedback

Facilities and Safety	• **SECURITY** – prevent unwanted risk and hazards
	• **CONTAINMENT** – minimize and keep away from community
	• **SAFETY** – policies and procedures; prevent harm
	• **ERGONOMICS** – safe and healthy working conditions, address facility and equipment adaptations
	  ○ less occurrence of hazard
	  ○ arrangement of specific things in the lab that prevents possibility of occurring hazard

- **Biosafety:** minimum health standards that will prevent you from acquiring infection
- **Biosecurity**: protect hazardous samples from theft; to prevent bioterrorism

INTERNATIONAL LABORATORY STANDARDS

- To be achieved by the laboratory; to know if it follows minimum protocol for specimen safety and specimen result quality
- Assess and Measure performance against International Quality Standards
- Laboratory Standard Organization:
  - ISO – International Organization for Standardization
    - Movement of papers and documents in the lab
    - Quality and competence of employee
    - Speed + accuracy in the lab
  - CLSI – Clinical and Laboratory Standards Institute *formerly*, National Committee for Clinical Laboratory Standards (NCCLS)
    - Specific performance in the lab

- ISO Standards: (Geneva)
  - ISO 15189: 2007
    - Medical Laboratories: Particular Req. for Quality and Competence
  - ISO/IEC 17025: 2005
    - General Req. for the Competence of Testing and Calibration

- CLSI Standards: (Wayne)
  - HSI-A2 (2004)
    - Quality Management System Model for Health Care, 2nd Ed.
    - Product: Path of Workflow
  - GP26-A3 (2004)
    - Application of a Quality Management System Model for Lab Services, 3rd Ed

II. FACILITIES & SAFETY

- Major Concern Person: **LABORATORY MANAGERS** (responsible for reinforcing safety in the lab)
- Responsible for Safety in the Laboratory
  - Laboratory Director: participate in planning and assessment
  - Quality Manager: conduct regular laboratory safety audits
  - Laboratorian: be aware and understand safety protocols; responsible for safety in the lab
- Laboratory Design
  - Most Efficient Design = Related Activities, Place in Close Proximity
  - Access: separate access of patient and staff
  - Circulation Pathways: collection, processing, post-exam
- Geographic Organization
  - Delineation of Laboratory Activities
  - Location of Service Rooms
  - Location of Activities with Specific Requirements:
    - Molecular Biology: separate, at least 2 rooms
      - Room 1: DNA Extraction (very sensitive to contamination; very sterile)
      - Room 2: Reagent and Amplification (organism is dead; it's ok not to wear face mask)
    - Fluorescence Microscopy: dark room, proper ventilation
    - UV Illumination for DNA gel photography: appropriate eye protection and dark room
- Spatial Provision for Equipment
  - Access to Equipment for **ENTRY** and **MAINTENANCE**
  - Power Supply: stable with backup
  - Liquid Disposal from Equipment: awareness and compliance
    - Ex: waste container with sensor
- Facilities and Benches
  - High Ceiling
  - Walls and Ceiling: painted with **GLOSSY PAINT**
  - Floor: easy to clean, no edge between walls and floor;
  - Work Benches:
    - Ceramic Tiles: good for benches
    - Grout: caution taken, may harbor contaminants
    - Steel: disadvantage: **RUST when cleansed with Chlorine**
  - Cleaning:
    - Weekly: Ceilings and Walls
    - Monthly: Storage areas, Refrigerators
- Procedures and Exercise
  - Fire Drills: Evacuation Procedures, Basic Security Precautions
- Waste Management
  - Wastes must be **COLOR CODED**
  - Special Attention: **BREAKAGE, SHARPS, NEEDLES**
  - Use **INTERNATIONALLY RECOGNIZED LABELS**

FIRST STEP IN SAFETY**: IDENTIFICATION OF HAZARDS**

- Types of Hazards:
  - Physical
  - Chemical and Radio Chemicals
  - Needles and Sharps
  - Biological
- Material Safety Data Sheet – available, kept close to hazardous material
- MAIN SOURCE OF CONTAMINATION - Specimen

LABORATORY ACQUIRED INFECTIONS:

DISEASE	PROBABLE SOURCE	MAXIMUM DISTANCE FROM SOURCE
Brucellosis	Centrifugation (cover)	Basement to 3rd Floor
Coccidioidomycosis	Culture transfer, Solid Media	2 building floors
Coxsackie Virus	Spilled tube of infected mouse tissue on floor	5 feet estimate
Murine Typhus	Intranasal Inoculation of Mice	6 feet estimate
Tularemia	20 petri-plates dropped	70 feet
Venezuelan Encephalitis	9 lyophilized ampoules dropped	4th floor stairs to 3rd or 5th floor

PERSONAL PROTECTIVE EQUIPMENT (PPE)

- Major Routes of Inoculation
  - Percutaneous
  - Contact between Mucus Membrane and Contaminated Material
  - Accidental Ingestion
- Access, Trainings, Habitual use (Self-Discipline)
- Types of PPE
  - Gloves: remove when leaving lab, never reuse (porous)
  - Masks: face shield glass, FFP2, US NIOSH-N95 (respirator) certified
  - Body Coat: required in Biosafety Level 2 laboratories, BSL3 is required to use a disposable coat

EMERGENCY MANAGEMENT

CHEMICAL SPILLS:

- **Minor Spill** – person is familiar with the chemical and its hazard
  - Recommended Steps:
    - Alert Co-workers, Clean Spill
      - Place a cloth on the spill and wet it with disinfectant (indirectly)
    - Free Liquid and Appropriate Absorbent
      - Caustic Liquid – Propylene Pad or Diatomaceous
      - Oxidizing Acid – Diatomaceous Earth
      - Mineral Acids – Baking Soda or Propylene Pads
      - Flammables – Propylene Pads, Sand

- **Major Spill –** beyond minor, needs expert assistance
  - Recommended Steps;
    - Alert Co-workers
    - Move to Safe Location
    - Call authorities to Report Situation

BIOLOGICAL SPILLS:

- Apply absorbent towel to Spill
- Disinfect Towel - 20 minutes
- Clean surface – Alcohol or soap and water or notify supervisor and other authorities
- Disinfectant:
  - Most Spills: **1:50 NaHOCl (50 g/L Chlorine)**
  - Large amount of Organics Spills: **1:10 (5g/L)**
  - Alcohol is NOT RECOMMENDED
    - Evaporates Quickly; Shorten Contact Time (should be 30 secs – 1 min)
    - 70% alcohol – due to water concentration

LABORATORY FIRE:

- Liquids with low flash points may ignite when near to heat
  - Ex. if sodium azide preservative is used, don't process near a metallic material = burst
- Small Lab Fire:
  - Extinguishable – if within 1-2 minutes
  - Cover fire with Inverted Beaker or Wet Paper Towel
  - If failed: **USE FIRE EXTINGUISHER**
  - Fire Extinguisher – CHECK ANNUALLY

III. EQUIPMENT

- Calibration and Maintenance – train everyone who uses it
- Day-to-Day maintenance – responsibility of technical operator
- Proper Selection of Equipment (consider the following):
  - Necessities, Performance of Machine, Required Cost, Reagents Readily Available, Easy Operation, Warranty and Maintenance, Safety Issues
- Equipment Installation:
  - Change package content contains all parts
  - Make a copy of software that is part of the system
  - Performance is verified and testing personnel are trained by the manufacturer

- Designate those who are authorized to use the specimen

- Post Installation:
  - Equipment Calibration
  - Performance Evaluation
  - Function Check
  - Preventive Maintenance
  - Maintenance Plan
  - Equipment Plan
  - Equipment Inventory
  - Inventory of Spare Parts
- Machine Operation
  - Trouble Shooting
  - Service and Repair
  - Retiring and Disposing
- Equipment Record of Maintenance
  - Documentation

IV. PURCHASING AND INVENTORY

- GOAL: Ensure AVAILABILITY QUANTITY; Prevent WASTAGE
- Appointed Responsible Person
  - Analyze Needs of the Lab
  - Establish minimum stock needed for a period
  - Appropriate Logs and Forms
  - Procedure for Receiving, Inspecting, Storing
- Effect of well Managed Inventory:
  - Uninterrupted Flow of Needed Materials
  - Ensure Availability of Products
  - Ensure Patient Needs are Met

V. PROCESS CONTROL

- SAMPLE MANAGEMENT
  - Handbook for Sample Management
    - Collecting, Processing, Testing Information
  - Systematic Sample Tracking within the Laboratory (F.I.F.O)
  - Established policy for specimen Storage and Disposal
  - Maintain Sample Integrity
  - Directly Affect Patient Care and Outcome
- QUALITY CONTROL
  - Used to monitor ANALTICAL PHASE
  - Goal: **ENSURE ACCURACY AND RELIABILITY**
  - Purpose: **DETECT, EVALUATE AND CORRRECT ERRORS** due to system failure environmental conditions
  - QC for varying methods:
    - Quantitative Exam: End-Point Value
    - Qualitative Exam: Presence or Absence
    - Semi-quantitative Exam: Expressed in Estimates
  - Poor Quality Control: Do not Report Result

VI. ASSESSMENT

- PURPOSE: Identify Problem, Improve Process and Procedure
- Find Root Cause and Take Corrective Actions

EXTERNAL AUDIT

- Done by groups or agencies outside the laboratory
- PURPOSE: **ACCREDITATION, CERTIFICATION, LICENSURE**
- Types of EQA:
  - Proficiency Testing
    - External provider of unknown sample
    - Cost effective, gives good, objective measure of laboratory testing
  - Rechecking or Retesting
    - Re-reading of slide by reference lab
    - Expensive and uses considerable time
  - On-Site Evaluation
    - Done when proficiency testing and rechecking are difficult to conduct
    - Can give *true picture* of lab's overall performance

INTERNAL AUDIT

- Done by staff in one area of the lab conduct of another area
- Provide quick information ON PERFORMANCE & COMPLIANCE
- Information Gathered in an Audit:
  - Processes and operating procedures
  - Staff competence and training
  - Equipment
  - Environment
  - Handling of samples
  - Quality control and verification of results
  - Recording and reporting practices
- Internal Auditor/ Quality Manager:
  - Develop a formal plan
  - Prepare a checklist based on selected guideline or standards
  - Meet with all staff and explain the audit process
  - Select staff to serve as auditors
  - Collect and analyze information
  - *Share results with staff*
  - Prepare a report
  - Present the report to management
  - Retain the report as a permanent laboratory record.
- Who can perform Internal Audit: **ANY KNOWLEDGEABLE STAFF**

ACCREDITATION

- Basic Terms:
  - Normative Document
    - Provides rules, guidelines, or characteristics and their results
    - Standards, Technical Specifications, Code of Practice, Regulation
  - Standard Document
    - Established by **CONCENSUS**
    - Approved by **RECOGNIZED BODY** that provides common and repeated use to achieve **OPTIMUM (QUALITY/ STANDARD) DEGREE OF ORDER** in a given context
  - Regulation
    - Standard that is mandated by a governmental agency or authoritative body
- Accrediting Body:
  - **ISO** - International Organization for Standardization
  - **CLIS** - Clinical Laboratory Standard Institute
  - **CEN** - European Committee for Standardization
  - **WHO** - World Health Authorization
- Types of Standardization:
  - Certification – Independent body gives "written assurance"
  - Accreditation – Authoritative body gives "formal recognition
  - Licensure – granting of ability to practice

VII. PERSONNEL

- Job Description – reflect all skills needed, accurately describe tasks, roles and authorities
- Competency of Personnel – evaluate at the time of hiring, and on regular, recurring basis
- Management Role:
  - **ATTRACT QUALIFIED PERSONNELS**
  - Provide motivation and appropriate benefits
  - Provide appropriate **WORKING CONDITIONS**
- Successful Personnel Management **= INCREASED RETENTION**
- Personnel Records:
  - Policy – Store information that is integral only to laboratory work-related information
- WHAT
  - Employment details
  - Original application and resume
  - Tests the employee is authorized to perform
  - Conditions of continued employment
  - Job description
  - Both original and subsequent competency assessments
  - Continuing education programs attended;
  - Personnel actions – corrective, disciplinary
  - Leave records
  - Health information, including records of work injury or exposure to occupational
  - Hazards, vaccine status, skin test (if any)
  - Performance appraisals
  - Emergency contact information
- WHERE
  - Secure Site to Protect Confidentiality

VIII. CUSTOMER SERVICE

- Customer satisfaction is one of the goals of the laboratory
- To address Customer Satisfaction:
  - Good Planning
  - Development of Appropriate Monitoring Tools (feedback forms)
  - Knowledge to apply the tools to Gain Useable Information
- Customers: Physician, Health Care Providers, Clinic Staff, Patients and their families, Public Health Officials, General Community
- Primary Resource: **STAFF TIME**
- "**TIP OF THE ICEBERG" -** may only be the received complaints
- Received complaints may not be the sole source of customer satisfaction assessment

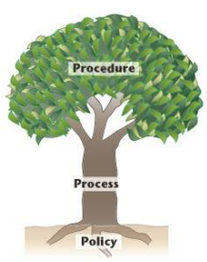
IX. OCCURRENCE MANAGEMENT

- Deal with Laboratory Errors
- Occurrence – any event that has negative impact on organization
- Occurrence Cycle:
  - Perform Investigation
  - Early Detect Occurrence
  - Root Cause Analysis
- Occurrence Management Process:
  - Establish process to detect problems
  - Keep a log of all problems: errors, date, time, action taken
  - Take Necessary Action – **REMEDIAL or CORRECTIVE**
  - Monitor and Observe for any RECURRENCE (if systematic)
  - Provide Information to:
    - All those who need it
    - All who are affected
- Keep good records – Problems, Investigations, Action Taken

X. PROCESS IMPROVEMENT

- Deming's **PLAN-DO-CHECK-ACT (PDCA)**
- New Tools:
  - Lean – Optimizing space, time, activity to improve physical path or workflow;
    IMPROVED LAB FLOOR PLAN
  - Six Sigma – Define, Measure, Analyze, Improve, Control
- Quality – **CONFORMANCE TO REQUIREMENTS**; not goodness or elegance
- Developing Successful Indicators:
  - Objective – measurable, not subjective
  - Methodology Available – available necessary tools
  - Limits – acceptable value; upper/lower limit
  - Interpretation – know in advance before anything
  - Limitation – what is being identified and not
  - Presentation – how to fully display the values
  - Action Plan – How, Who, & How long to collect data
  - EXIT Plan – when to STOP, replace if stable/working
- **Characteristics of Good Indicators** (page 176)
  - Measurable: the evidence can be gathered and counted;
  - Achievable: the laboratory has the capability of gathering the evidence it needs;
  - Interpretable: once it is gathered, the laboratory can make a conclusion about the information that is useful to the laboratory;
  - Actionable: if the indicator information reports a high or unacceptable level of error, it is possible to do something about the problem identified;
  - Balanced: consider indicators that examine multiple aspects of the total testing cycle in the pre-examination, examination, and post-examination phases;
  - Engaging: indicators should examine the work of all staff, not just one group;
  - Timed: consider indicators with both short-term and long-term implications.
- MOST IMPORTANT – **GENERATE SOLUTIONS**

XI. DOCUMENTS AND RECORDS



- Procedures: should be seen visually
- Processes: only the staff of the laboratory knows
- Policies: CMT, lab managers, section heads, pathologist/ administrative positions know
- Document Hierarchy
- Importance of Document Keeping:
  - Quality Manual – overall guiding document for quality system
  - SOPs - contains step-by-step instructions; ensure consistence
    - CRUSH OUT ONCE, put the date, and affix signature; If there is a mistake.
  - Reference Material – good scientific & clinical info about disease
- DO WHAT YOU WROTE AND FOLLOW WHAT YOU ARE WRITING
- Accessibility – accessible to staff and laboratory person only with direct relation to the process

RECORDS

- Printed (hand or computer); NOT REVISED or MODIFIED
- Complete, Legible, Carefully Maintained
- Uses:
  - Continuous Monitoring
  - Tracking of Samples
  - Evaluation of Problems
  - Management
- Never Change the Record – not addition (date, signature or initials)
- **Storage System:**

	PAPER SYSTEM	ELECTRONIC SYSTEM
Permanence	Permanent ink, with Pages (Table of Contents)	Needs good BACK UP SYSTEM Regularly Maintained
Accessibility	Poor retrieval	Good Retrieval
Security	People within the lab	People within the lab
	Keep from SPILLS	
Traceability	GOOD IN TERMS OF Chronological updates with signature	Poor if user is not knowledgeable enough

- Ensure Good Retention of Records

12. INFORMATION MANAGEMENT

- Include Identifier:
  - **Patient Identifiers**
  - **Sample Identifiers**
- One Data = One Unique Identifier
- Information Management
  - System that incorporates all the processes needed for effectively managing data – both incoming and outgoing patient information
- Effects of Good Information Management;
  - Ensure that all data – the final product of laboratory – is well managed
  - Consider all the ways laboratory data will be used when planning a system
  - Ensure the accessibility, accuracy, timeliness and security of data
  - Ensure confidentiality and privacy of information

XIII. ORGANIZATION

- Management and the supporting organizational structure of the laboratory
- Function of Organization: **IMPLEMENTATION**
  - Commit
  - Prepare to Implement in Stages
  - Determine Resource Requirements
  - Engage all staff
- Roles:
  - Provide resources
  - Monitor basics
  - Establish Monitoring Program
- Organizational Steps
  - Assign responsibility for implementation
  - Allocate resources
  - Develop and Distribute Quality Manual
  - Begin Implementation
  - Monitor Compliance
- Successful Implementation
  - Planning
  - Management Commitment
  - Understanding the Benefits
  - Engaging Staff at all levels
  - Settling Realistic Time Frame
  - Looking for Ways to Continually Improve

LABORATORY OPERATIONS

PHASES OF ANALYSIS	
Pre-Analytical	• Test order by physician, test order by nursing/clerical staff, patient preparation, patient ID, specimen collection, specimen transport, ***specimen processing***
Analytical	• Test Analysis, QC, reagents, calibration, preventive maintenance
Post-Analytical	• Verifications of ***calculations and reference ranges***, review of results, procedures for notification of critical values, result reporting, test interpretation by physician, follow-up patient care

QUALITY ASSURANCE vs. QUALITY CONTROL	
QUALITY ASSURANCE (acceptable error: <1%)	**QUALITY CONTROL**
Pre-Analytical, Analytical, Post-Analytical	Analytical

SPECIAL SITUATIONS IN PHLEBOTOMY	
IV	• Use opposite arm or perform fingerstick, if possible; otherwise, have nurse turn off IV for *2 min*., applying tourniquet below IV, use different vein (if possible). • Document location of IV & venipuncture, type of fluid
Fistula	• Draw from opposite arm
Indwelling lines & catheters, heparin locks, cannulas	• Usually not drawn by lab • First 5 mL drawn should be discarded. Lab may draw below heparin lock if nothing is being infused
Sclerosed veins	• Select another site
Hematoma	• Draw below
Streptokinase/ tissue plasminogen activator (TPA)	• Minimize venipunctures. Hold pressure until bleeding has stopped
Edema, Scar, Burns, Tattoos	• Select another site • Median cubital = preferred site; most stable vein (well anchored by the cephalic and basilic vein)
Mastectomy	• Draw from opposite arm
Patient Refuses	• Try to persuade. If unsuccessful, notify the nurse. • *Never draw without consent;* could lead to charges of assault or battery
Unidentified Patient	• Ask nurse to ID before drawing

SPECIAL TEST REQUIREMENTS		
REQUIREMENT	**EXAMPLES**	**NOTES**
Fasting	• FBS, TAG, Lipid Panel, Gastrin, Insulin	• Nothing to eat or drink (except water) for at least 8 hours
Chilling	• ACTH, Acetone, Ammonia, Gastrin, Glucagon, Lactic Acid, Pyruvate, PTH, Renin	• Place in slurry or crushed ice & water. • Don't use ice cubes alone because RBCs may lyse
Warming	• Cold Agglutinins, Cryoglobulins	• Use 37°C heat block, heel warmer, or hold in hand
Protection from Light	• Bilirubin, Carotene, RBC, Protoporphyrin, Vitamin A & B12	• Wrap in Aluminum Foil • Photo-oxidation – when exposed to light
Chain of Custody	• Any test as evidence in legal proceedings; e.g. blood alcohol, drug screen, DNA Analysis	• Chain of custody form. • Lock Box may be required

*Follow Laboratory's Established Procedures

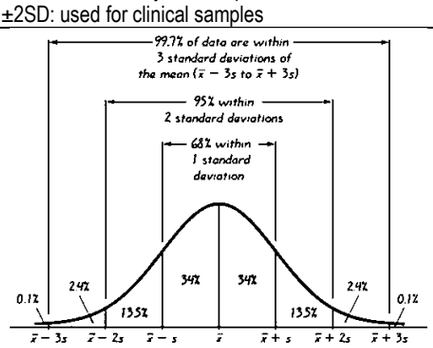
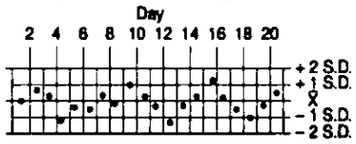
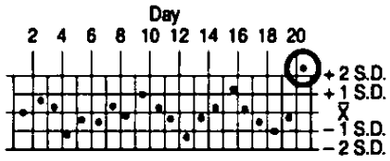
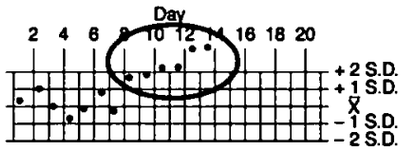
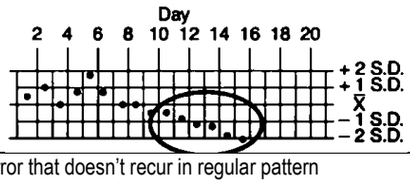
PHLEBOTOMY SOURCES OF ERROR	
Misidentification of patient	• Treatment errors, possibility of Transfusion Fatality
Drawing at Incorrect Time	• Treatment errors (TDM, Analytes exhibiting diurnal variations)
Improper skin disinfection	• Infection at site of puncture. Contamination of Blood C/S and component. • Alcohol contamination in blood alcohol level (False Increase)
Drawing from Edematous Site	• Sample dilution with Tissue Fluid
Fist Pumping during venipuncture (Hemolysis)	• INCREASED: K^+, Lactic Acid, Ca^{2+}, Phosphorus • DECREASED: pH
Tourniquet > 1min. (Hemoconcentration)	• INCREASED: K^+, Total Protein, Lactic Acid
IV Fluid contamination	• INCREASED: Glucose, Electrolytes (Depending on IV)
Expired Collection tubes	• DECREASED: Vacuum, Failure to obtain specimen
Incorrect Anticoagulant or Order of Draw	• K_2 EDTA before serum or heparin tube: $\downarrow Ca^{2+}$, Mg^{2+}, $\uparrow K^+$ • Contamination of Citrate Tube with clot activator: Erroneous Coagulation
Failure to Hold Bottom of tube lower than top during collection	• Carryover • Additive Contamination
Short Draw	• Incorrect Blood:Anticoagulant Ratio (e.g. Coagulation test)
Inadequate mixing of anticoagulant	• Micro-Clots, Fibrin, Platelet Clumping
Hemolysis form Alcohol Contamination	• $\uparrow K^+$, Mg^{2+}, LD, Iron also "Milking" Site of Capillary Puncture, Probing with Needle, Vigorous Shaking of Tubes, Exposure of Samples to Extremes of Temperature

GUIDELINES FOR SPECIMEN HANDLING AND PROCESSING

1. Careful Blood Transport – avoid hemolysis
2. Maintain tubes in VERTICAL POSITION – promote complete clotting
3. Allow serum & gel separator tubes to clot for 30-60 min. before centrifugation to avoid fibrin strands
4. Centrifuge within 2 hours of collection. Separate serum or plasma from cells within 2 hrs. of collection (exception: centrifuged gel tubes)
5. Spin ***most tubes*** at 1,000-1,300 RCF for 10-15 mins.
6. Spin ***Citrate tubes*** at 1,500 RCF for 15 mins. To produce platelet-poor plasma
7. Keep tubes capped during centrifugation to avoid loss of CO_2, change of pH, evaporation or aerosol formation
8. Don't re-spin primary tubes. Can cause hemolysis. If re-centrifuging is necessary, transfer serum/plasma to another tubes
9. Don't re-spin SST. Serum in contact with RBCs under gel can be expressed &$\uparrow K^+$
10. When transferring samples to secondary containers, aspirate to avoid cellular contamination. Don't pour
11. Lipemic specimens can be ultracentrifuged at 105 x g to remove chylomicrons (triglycerides)
12. Separated serum/plasma may be kept at RT for 8 hrs. or at 2-8°C for 48 hrs. For longer storage, freeze at 20°C. Avoid repeat freezing & thawing
13. Don't freeze whole blood (RBC = consumes nutrients)

QUALITY CONTROL	
QA	• Lab ensures quality results – closely monitor test analytical stage
QC	• Part of analytical phases; monitor results from control samples • For machine and reagents
Control	• Chemically & Physically similar samples to unknown specimen & is tested in exactly the same manner • Purpose: *Monitors precision, CLIA requires at least 2 levels of controls each day test is performed*
External QC	• Controls NOT BUILT into test system. • e.g. Proficiency Testing
Internal Monitoring	• Controls that ate BUILT into test system

QUALITY CONTROL STATISTICS	
Measure of Dispersion	• Statistical parameters describing spread of data about mean, • Example: Standard Deviation (SD), Coefficient of Variation (CV), Range, Measurements of Precision
Range	• Difference between highest & lowest values in data set
Mean	• Sum of all observations divided by number of observations
Standard Deviation	• Statistical expression of dispersion of values around mean. • Requires a minimum of 20 VALUES $s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$ Where: • Σ = sum • x = individual value • x̄ = mean • n = number of values
Coefficient of Variation (CV)	• Expresses standard deviation as percentage. • Formula: CV% = (SD ÷ mean) x 100 • The ↓ the CV, the ↑ the Precision

NORMAL GAUSSIAN DISTRIBUTION CURVE	
Normal Distribution	• The Gaussian bell-shaped curve: ○ 68% of values fall w/in ± 1 SD of mean ○ 95% of values fall w/in ± 2 SD of mean ○ 99.7% of values fall w/in ± 3 SD of mean • CV = 68%, 95%, 99.7% • The lesser the CV, the better the precision • Most precise: ± 1 SD, CV is 68% • If it has exceeded 99.7% or ±3SD, it means that there is a problem in the machine • ±3SD: used for any SD computation • ±2SD: used for clinical samples
Gaussian Curve	 • Normally observed / only occur when mean = mode = median • Gaussian frequency distribution with 95% limits indicated.
QUALITY CONTROL STATISTICS	
Control Limits	• Range where control values must fall to consider test valid • Commonly used mean (clinical samples): ± 2SD • 1 determination in 20 will fall outside ± 2SD - anticipated part of normal variation ○ accepted: 1 SD exceeded ± 2SD ○ not accepted: >1 exceeded ± 2SD  • Levey-Jennings Chart showing in-control results
Levey-Jennings Chart	• Normal distribution curve lying on its side, marked with mean, ± 1SD, ± 2SD, ± 3SD • Used to plot SD computed in the previous days
Outlier	• Definition: A control result outside established limits • Cause: Chance or test system problem  • If it occurs more than once in 20 successive runs, investigation must be carried out.
Shift	• 6 consecutive control values on same side of mean 
Trend	• Control values increasing or decreasing for 6 consecutive runs. 
Random Error	• Error that doesn't recur in regular pattern • Most common
	• Causes: ○ dirty glassware ○ wrong pipette ○ voltage fluctuation ○ sampling error ○ Anticoagulant / drug interference
	• Westgard Rule: 1_{3s} or R_{4s} • Usually ONE-TIME ERROR (if it occurs more than once, it is not a random error)
	• Remedy: controls and samples can be rerun with success
Systematic Error	• Recurring error inherent in test procedure
	• Causes: ○ dirty photometer ○ faulty ISE ○ evaporation/contamination of standards or reagents
	• Westgard Rule: affects all results 2_{2s}, 4_{1s} or 10_x
	• Remedy: requires investigation to determine cause
False Rejection	• Rejection of run because QC results indicate problems when none is present. • Use of Westgard rules minimizes false rejections

WESTGARD CONTROL RULE

RULE	EXPLANATION	TYPE OF ERROR	COMMENTS
1_{2s}	1 control >±2s from mean. Warning flag of possible change in accuracy or precision.	WARNING	Initiates testing of other rules. If no violation of other rules, run is considered in control.
1_{3s}	1 control >±3s from mean	Random	Rejection Rule
2_{2s}	2 consecutive controls >2s from mean on the same side	Systematic	Rejection Rule
R_{4s}	2 consecutive controls differ by 4s	Random	Rejection Rule
4_{1s}	4 consecutive controls >1s from mean on same side	Systematic	Rejection Rule
10_x	10 consecutive controls on same side of mean	Systematic	Rejection Rule

* ALL systematic are even numbers: **2_{2s}, 4_{1s}, 10_x**

TYPICAL STEPS TAKEN WHEN CONTROL IS OUT OF ACCEPTABLE RANGE

STEP	REASON
1. Hold until problem is solved	• Inaccurate control = inaccurate result
2. Rerun control once	• Maybe due to expected random error • 1 in 20 results will be outside ±2SD
3. Unresolved? ○ Run a new vial of control or another lot number	• Control might have been outdated, improperly stored, contaminated
4. Unresolved? ○ Look for & correct any problems, then run control	• Consider reagents (low, outdated, improperly stored, contaminated, change in lot number), preventive maintenance (overdue), mechanical problems, clots, etc.
5. Unresolved? ○ Recalibrate, then run control	• Calibration may have shifted
6. Unresolved? ○ Get assistance!	• Supervisor or service rep may be able to determine problem
7. Resolved? ○ Document corrective action	• Provides a record for future reference, points out repetitive problems
8. Repeat tests & issue corrected reports, as needed	• Ensure accuracy of reported results

CALIBRATION	
Calibration	• Testing & Adjusting analyzer's readout • Establish correlation between measured & actual concentrations
Calibrator	• Reference material with known concentration of analyte. • Formerly known as **Standard** • Mixture of different analytes ○ Standard = colorless ○ Control = colored-like-serum
Calibration verification	• Test 3 Levels = **High, Normal, Low** • Required every 6 months, when lot # of reagents changes, following preventive maintenance or repair, & when controls are out of range • Recalibrate/recondition the machine

POINT-OF-CARE TESTING	
POCT	• Testing performed at site of patient care. • a.k.a. **Decentralized**, Bedside, or Near-Patient Testing
Goal	• Provide **rapid test results** where immediate medical action is required, e.g., emergency department, intensive care units.
Common POCT tests	• Urine reagent strips, glucose, electrolytes, blood gases, activated clotting time (ACT), PT, APTT, hemoglobin.
Optimal staffing	• **Point-of-Care Coordinator (POCC)** ○ Oversees testing & responsible for compliance, training, QC, proficiency testing (if required) • **Testing Personnel:** ○ Qualifications determined by test complexity of testing. Usually phlebotomists, lab assistants, nurses, respiratory therapists.
Considerations	• Cost, Specifications, Ease of Use, TAT, Quality & , data management/connectivity/interface capabilities.

CREDENTIALING	
Accreditation	• Institution; Meet certain Standards; Voluntary.
Certification	• Voluntary; Person; Meet Education Requirements & Demonstrate Entry-Level Competency by Passing Exam.
Licensure	• Permission to engage in certain professions/businesses. • Mandatory, Illegal to practice/operate in that state without license

AGENCIES FOR COMPLIANCE AND REGULATIONS	
AABB	• Blood Bank Standards and Accreditation • *Association for the Advancement of Blood & Biotherapies* • *Formerly: American Association of Blood Banks*
CDC	• Infection Control & Safe Work Practices; Standard & Guidelines • *Centers for Disease Control and Prevention*
CLSI	• Standards on All Aspects of Lab Practice • *Clinical and Laboratory Standards Institute*
ISO	• Standard to facilitate international exchange of goods & services. • *International Organization for Standardization*

FEDERAL REGULATIONS	
Hazard Communication Standard (OSHA 1983) *"Right to Know Law"*	• Requires employers to inform employees about hazards & educate in safe handling.
Clinical Laboratory Improvement Amendments of 1988 *"CLIA '88"*	• Regulates all lab testing (except research) performed on humans in U.S administered by CMS.
Occupational Exposure to Hazardous Chemicals in Laboratories (OSHA 1990) *"Laboratory Standard"*	• Requires Chemical Hygiene Plan to minimize personnel exposure to hazardous chemicals in labs.
Bloodborne Pathogens Standard (OSHA 1991)	• Mandates work practices & procedures to minimize worker exposure to bloodborne pathogens.
Formaldehyde Standard (OSHA 1992)	• Requires monitoring of formaldehyde exposure
Health Insurance Portability and Accountability ACT of 1996 "HIPAA"	• Regulates use & disclosure of protected health information (PHI)

SAFETY IN THE LABORATORY

SAFETY PROGRAMS AND PRACTICES	
A. BLOOD BORNE PATHOGEN STANDARD	
Purpose	To protect health-care workers from occupational exposure to bloodborne pathogens (BBP; e.g., HIV, HBV, HCV)

Primary Requirements in Blood Borne Standard:

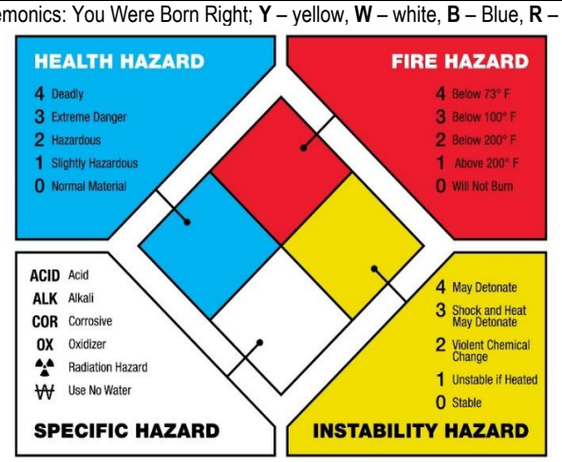
1. Exposure Control Plan
2. Universal Precautions
3. Engineering Controls
4. Work Practice Controls (order = manner that decreases hazard)
5. Personal Protective Clothing & Equipment
6. Housekeeping
7. Training
8. Medical Surveillance
9. Hepatitis B vaccine
10. Hazard Communication
11. Sharps Injury Log

B. HAZARD COMMUNICATION STANDARD	
Aka	"Right to Know" Law; "HAZCOM"
Purpose	INFORM employees on chemical hazards in workplace & protective measures
Primary Requirements	1. Written hazard communication plan 2. Inventory of hazardous chemicals on site 3. Hazard labeling 4. Material safety data sheet (MSDS) for each chemical readily accessible to employees on each shift. 5. Training on initial assignment & when new hazard introduced.
C. CHEMICAL HYGIENE STANDARD	
Aka	Laboratory Standard
Purpose	LIMIT employee exposure to hazardous chemicals to levels at or below *permissible exposure levels (PELs)*
Primary Requirements	1. *Written* chemical hygiene plan 2. Designation of chemical hygiene officer 3. Hazard identification & labeling 4. Material safety data sheet (MSDS) for each chemical readily accessible to employees on each shift. 5. Use of personal protective equipment 6. Proper maintenance of fume hoods & other PPE 7. Monitoring of employee exposure to hazardous chemicals 8. Medical exams at no cost in cases of suspected overexposure 9. Training on initial assignment & before assignments

HAZARD CATEGORIES OF CHEMICALS	
CLASSIFICATION	**EFFECT**
Corrosives	• Tissues Destruction (visible) by exposure (contact/Inhalation)
Toxic Substances	• Interfere Metabolism (Ingestion, Inhalation, Absorption, through Intestine)
Carcinogens	• Capable of Causing Cancer
Mutagens & Teratogens	• Induce Genetic Mutation • Cause Defect in Embryo
Ignitable/s	• Fire
Reactive/s	• Explosions

NFPA HAZMAT DIAMOND

(Mnemonics: You Were Born Right; **Y** – yellow, **W** – white, **B** – Blue, **R** – Red)



HAZARD	HEALTH	FLAMMABILITY
COLOR	Blue (Left)	Red (Top)
0	No Hazard	Will not Burn
1	Significant Irritation	Pre-heated for ignition to occur
2	Temporary Incapacitation	Heated or increase ambient temp. to burn
3	Serious of permanent injury	Ignited in Almost all Temperatures
4	Lethal	Will vaporize & burn at normal temp

HAZARD	INSTABILITY	SPECIAL HAZARD
COLOR	Yellow (Right)	White (Bottom)
0	Stable	**W** = unusual reactivity with water **OX** = oxidizer
1	Increase Temp. makes unstable	
2	Violent chemical change at increase temp. or pressures	
3	Explode from increase temp. or shock	
4	Explode at normal temp & pressures	

FIRE SAFETY		
CLASS	**COMBUSTIBLE**	**EXTINGUISHER**
A	Cloth, wood, paper	Pressurize water (A) Dry chemicals (A,B,C)
B *"Basa"*	Flammable/ Combustible Liquids	Dry chemical (ABC) CO_2 (BC)
C *"Curyente"*	Electrical Equipment	Dry chemical (ABC) CO_2 (BC)
D	Combustible Metals	Leave to professional firefighters
E	Arsenal Fire	Do not extinguish

MATERIAL SAFETY DATA SHEET (MSDS)	
Content	1. Health effects of exposure to the product 2. Hazard evaluation related to the product's handling, storage, or use 3. Measure to protect workers at risk of exposure emergency procedures
Purpose	• Supplement the alert information provided on labels
Every 3 yrs.	• MSDS Required Revision Period

PERSONAL PROTECTIVE EQUIPMENT (PPE)	
Application	• Shall be provided, used, and maintained in a sanitary and reliable condition *wherever it is encountered in a manner capable of causing injury* or impairment in the function of any part of the body through absorption, inhalation, or physical contact
PPE Provider	• Employer: assure adequacy, maintenance, sanitation
Hazard Assessment	• Identify which type of PPE shall be used
Employer Responsibilities	• Assess, Provide, Train, and re-train, maintain
NOT Employer Responsibilities	• Steel boots, everyday clothing, weather protection (sun screen, raincoat)

EMERGENCY PRACTICES	
Needle Stick Injury	• Post-exposure prophylaxis (PEP), Testing of Patient Samples
Fire Emergency	• Cover Fire with Inverted Beaker or Wet paper towel • If NOT EFFECTIVE: Use FIRE EXTINGUISHER (annually checked) • RACE – Rescue, Alarm, Contain, Extinguish • PASS – Pull, Aim, Squeeze, Sweep • Extinguishable Fire: if w/in 1-2 minutes Fire Out
Chemical Spill	• Minor Spill – person is familiar with chemical and its hazard • Major Spill – needs expert assistance
Biological Spill	• Apply Absorbent towel to spill • Disinfect Towel – 20 minutes • Clean Surface – alcohol or soap and water or Notify supervisor and other authorities • Disinfectant: • Most Spills: 1:50 NaHOCl (50g/L Chlorine) • Large Amount of Organic spills: 1:10 (5g/L) • Alcohol is NOT RECOMMENDED (evaporates quickly shorten contact time)

PACKAGING AND TRANSPORT OF SPECIMEN/MICROORGANISM	
Primary Container	• Test tube, vial, etc. containing etiologic agent. • Must be securely closed, watertight, surrounded by absorbent material & placed in secondary container
Secondary Container	• Must be watertight, sealed & placed in approved mailing container
Mailing Container	• Must be made of fiberboard
Labeling	• Biohazard label required on primary & mailing containers
Training	• Employees must be trained & retrained every 2-3 yrs. or when regulations change